

Cubic Base Case Report

Polynomial

$$f(q) = q^3$$

Naive

$$g_{\text{naive}}(X) = X^3$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 6\frac{\Delta^2}{\epsilon}q + q^3$$

$$\text{Var}(g_{\text{naive}}(X)) = 720\frac{\Delta^6}{\epsilon} + 324\frac{\Delta^4}{\epsilon}q^2 + 18\frac{\Delta^2}{\epsilon}q^4$$

$$\text{MSE}(g_{\text{naive}}) = 720\frac{\Delta^6}{\epsilon} + 360\frac{\Delta^4}{\epsilon}q^2 + 18\frac{\Delta^2}{\epsilon}q^4$$

$$\text{Bias}(g_{\text{naive}}) = 6\frac{\Delta^2}{\epsilon}q$$

Unbiased

$$g_{\text{unb}}(X) = X^3 - 6X\frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = q^3$$

$$\text{Var}(g_{\text{unb}}(X)) = 504\frac{\Delta^6}{\epsilon} + 252\frac{\Delta^4}{\epsilon}q^2 + 18\frac{\Delta^2}{\epsilon}q^4$$

$$\text{MSE}(g_{\text{unb}}) = 504\frac{\Delta^6}{\epsilon} + 252\frac{\Delta^4}{\epsilon}q^2 + 18\frac{\Delta^2}{\epsilon}q^4$$

Comparison

$$\text{mean gap} = -6 \frac{\Delta^2}{\epsilon} q$$

$$\text{variance gap} = -216 \frac{\Delta^6}{\epsilon} - 72 \frac{\Delta^4}{\epsilon} q^2$$

$$\text{MSE gap} = -216 \frac{\Delta^6}{\epsilon} - 108 \frac{\Delta^4}{\epsilon} q^2$$

$$\text{Bias}(g_{\text{naive}})^2 = 36 \frac{\Delta^4}{\epsilon} q^2$$